

Week 10: Self-Adjointness

The One Step

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Abstract

Week 09 [1] ended at the gap: the resonance object $(\mathcal{A}, \mathcal{H}, D)$ on the adèle class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$ is correctly identified, its spectrum is the zeros of $\zeta(s)$, and one step remains: show that D is self-adjoint on $\mathcal{H}$. Self-adjointness forces real spectrum. Real spectrum on this system, combined with the functional equation, forces $\text{Re}(\rho) = \frac{1}{2}$. That is the Riemann Hypothesis. This paper is the one step. It is given freely. It does not complete the proof. It carries the proof as far as it will go and names, one final time, what remains.

1 The Setup

Definition 1 (The operator). Let $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$ be the adèle class space, $\mathcal{H} = L^2(X, d^*x)$ the Hilbert space of square-integrable functions on X with respect to the multiplicative Haar measure d^*x , and U the unitary representation of $\mathbb{R}_+^*$ on $\mathcal{H}$ given by scaling:

$$(U(\lambda)f)(x) = f(\lambda^{-1}x), \quad \lambda \in \mathbb{R}_+^*.$$

The generator of U is the operator D :

$$U(\lambda) = \lambda^{iD}, \quad D = \frac{1}{i} \frac{d}{d\lambda} \Big|_{\lambda=1} U(\lambda).$$

D is formally self-adjoint: $\langle Df, g \rangle = \langle f, Dg \rangle$ for all f, g in the domain of D . The question is whether D is *essentially* self-adjoint: whether its closure $\bar{D}$ is self-adjoint, i.e., whether $\bar{D} = \bar{D}^*$.

2 What Self-Adjointness Would Give

Proposition 2 (Self-adjointness implies RH). *If D is self-adjoint on $\mathcal{H}$, then:*

1. *By the spectral theorem, D has real spectrum: all eigenvalues λ satisfy $\lambda \in \mathbb{R}$.*
2. *The zeros ρ of $\zeta(s)$ appear in the spectrum of D via $\rho = \frac{1}{2} + i\lambda$ (the identification of zeros with spectral parameters).*

3. Real λ implies $\rho = \frac{1}{2} + i\lambda$ has $\text{Re}(\rho) = \frac{1}{2}$.

4. Therefore all non-trivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = \frac{1}{2}$.

This is the Riemann Hypothesis. Self-adjointness of $D \Rightarrow RH$.

3 The Deficiency Indices

Definition 3 (Essential self-adjointness). A symmetric operator D on a Hilbert space $\mathcal{H}$ is *essentially self-adjoint* if and only if its deficiency indices are equal: $n_+ = n_-$, where

$$n_{\pm} = \dim \ker(D^* \mp i).$$

If $n_+ = n_- = 0$: D is essentially self-adjoint. Its closure is self-adjoint. The spectrum is real.

If $n_+ = n_- \neq 0$: D has self-adjoint extensions (parametrized by $U(n_+)$), but is not essentially self-adjoint. The choice of extension determines the spectrum.

If $n_+ \neq n_-$: D has no self-adjoint extension. The operator cannot be made self-adjoint.

Conjecture 4 (The deficiency conjecture). *The operator D on $\mathcal{H} = L^2(X, d^*x)$ has deficiency indices $n_+ = n_- = 0$. D is essentially self-adjoint. Its unique self-adjoint closure has real spectrum. The zeros of $\zeta(s)$ lie on $\text{Re}(s) = \frac{1}{2}$.*

4 What Is Known

Proposition 5 (The current state). *The following is established:*

1. *The operator $U(\lambda)$ is unitary on $\mathcal{H}$ for each $\lambda \in \mathbb{R}_+^*$ (Connes [2]).*
2. *D is formally symmetric: $D \subseteq D^*$ (formal computation).*
3. *The trace formula $\text{Tr}(f \cdot U) = W(h_f)$ reproduces the Weil explicit formula (Connes [2]).*
4. *The zeros of $\zeta(s)$ are absorption frequencies of D (spectral interpretation, Connes).*
5. *$W(h) \geq 0$ for test functions h supported on $[1, \infty)$ (partial positivity, known).*

What is not established:

- (6) *The deficiency indices of D are $(0, 0)$.*
- (7) *D is essentially self-adjoint.*
- (8) *$W(h) \geq 0$ for all admissible $h \geq 0$.*

Items (6), (7), (8) are equivalent. Each implies the Riemann Hypothesis. None is proved.

5 The One Step

Remark 6 (What a proof would do). To prove Conjecture 4, one must show that the equation

$$D^*f = \pm if$$

has no nonzero solution $f \in \mathcal{H}$. Equivalently: the operators $D^* \mp i$ have trivial kernel in $\mathcal{H}$.

The equation $D^*f = if$ in $L^2(X, d^*x)$ is a first-order differential equation on the adèle class space. Its solutions, if they exist, would be eigenfunctions of D^* with eigenvalue $\pm i$ — which would mean the spectrum of D is not real, which would mean $\zeta(s)$ has a zero off the critical line.

To show no such solution exists in $\mathcal{H}$: show that any solution to $D^*f = if$ grows faster than L^2 at infinity in X . This is a growth estimate on the adèle class space.

The Arithmetic Wall [3] is here: the estimate requires controlling the behavior of f at the archimedean place ∞ — the place that $\text{Spec } \mathbb{Z}$ is missing, the point that the Arakelov compactification adds but cannot animate. The archimedean control is the missing Frobenius, one final time, in functional-analytic dress.

The one step is the archimedean estimate. The Wall is archimedean. It has always been archimedean.

Theorem 7 (The final shape). *The Riemann Hypothesis is equivalent to:*

$$\text{Every solution } f \in \ker(D^* - i) \text{ on } L^2(X, d^*x) \text{ satisfies } f = 0.$$

This is a statement about L^2 growth on the adèle class space. It is a statement about the archimedean place. It is a statement about what the archimedean fiber of $\text{Spec } \mathbb{Z}$ can support.

The proof would show: the archimedean place cannot support an L^2 eigenfunction of D^ with eigenvalue i . This is the self-inhibiting mechanism [4]: the system rejects the eigenfunction. The eigenvalue i has no L^2 home in X . The zeros have no home off the line. They stay because there is nowhere else for them to be.*

Remark 8 (Given freely). This paper does not prove the Riemann Hypothesis. It carries the proof to the last door and names what is behind it: an L^2 estimate on the adèle class space, archimedean in nature, equivalent to the self-adjointness of D , equivalent to RH.

The estimate is not made here. The door is named. The shape is held.

Whoever opens it will find the zeros already on the line, the lever already still, the beam already going.

They were always there. We were learning to see them.

References

- [1] Rincón, D. (2026). Week 09: The Resonance Object. Preprint.
- [2] Rincón, D. (2026). Noncommutative Geometry and the Riemann Hypothesis. Preprint.
- [3] Rincón, D. (2026). The Arithmetic Wall. Preprint.

- [4] Rincón, D. (2026). Inhibition: The Second Function of the Aura. Preprint.
- [5] Rincón, D. (2026). Mathematics. Preprint.
- [6] Rincón, D. (2026). Laser Space. Preprint.